

ATLAS Overview

Dr. Christina Liaghati

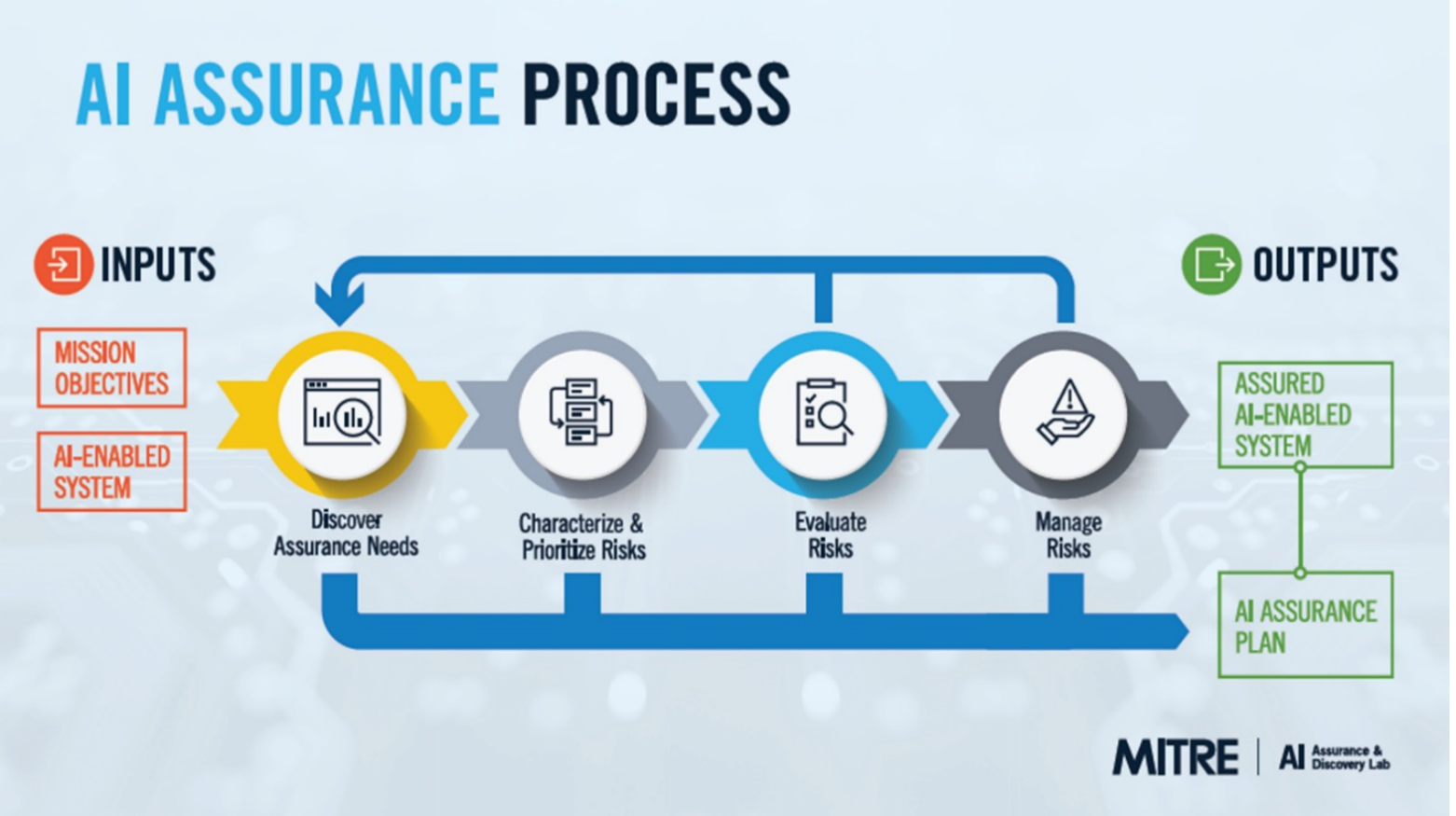
MITRE ATLAS Lead

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AI Assurance

AI assurance is a process for **discovering, assessing, and managing risk** throughout the lifecycle of an AI-enabled system so that it operates effectively to the benefit of its stakeholders.



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AI Intersections with Cyber

Security & Assurance of AI-enabled Systems

Securing the unique system vulnerabilities of AI-enabled systems – includes red teaming to discover vulnerabilities

Using AI in Offensive or Malicious Cyber Attacks

Attackers using AI in offensive assaults on both traditional cyber systems and AI-enabled systems

Using AI in Cybersecurity Practices

Using AI to improve our cybersecurity practices, i.e., detection, risk analysis, and defensive or mitigation techniques

MITRE has capabilities and teams working in all three areas
Today we are focusing on the security of AI-enabled systems

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Focusing on Real-World Demonstrated AI Attack Vectors

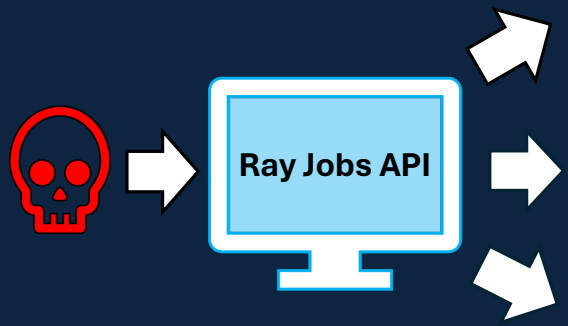
Incident - Malicious Attack by an Adversary

Ongoing real-world AI supply chain attack vector with estimated financial impact over $1 Billion (as of March 2024, that $ estimate would likely be much higher now)

ShadowRay: The lack of authorization in the Ray Jobs API default configuration allows adversaries to invoke arbitrary jobs on the API. This grants access to user tokens/PII and fraudulent use of cloud compute time at the cost of the user.

CVE-2023-48022





User PII
User Passwords
Slack Tokens
Slack Messages
Private SSH Keys

Ray Leaks
AI Production Workloads
Production DB Credentials
Full Ray Database Access

External Services
OpenAI Tokens
KubernetesAPI Access
Stripe Tokens
HuggingFace Tokens

ShadowRay: First Known Attack Campaign Targeting AI Workloads Exploited In The Wild | Oligo Security

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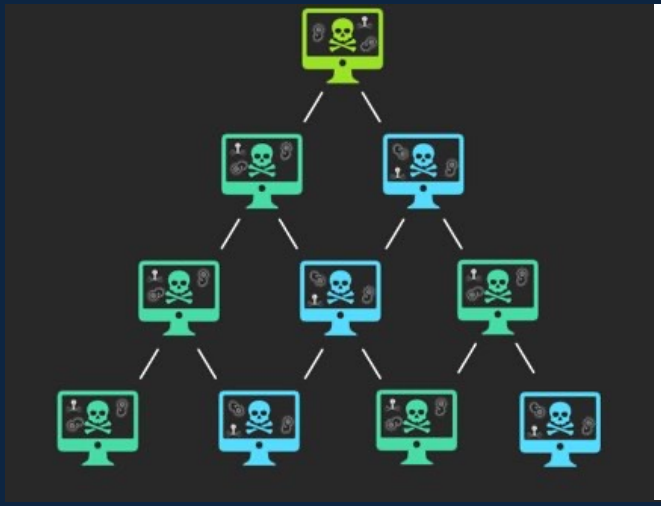
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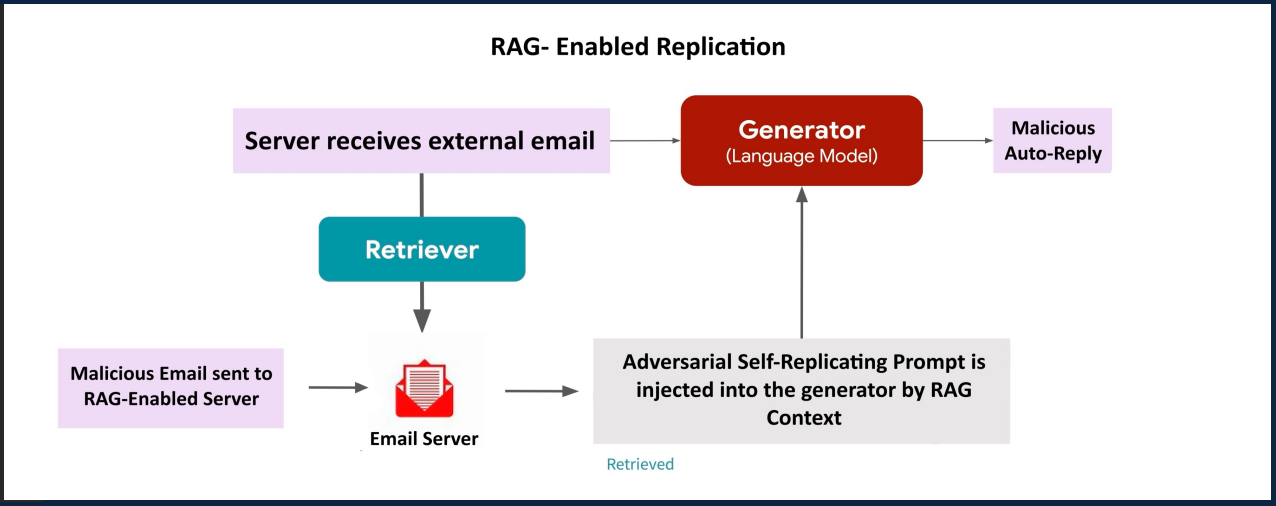
Focusing on Real-World Demonstrated AI Attack Vectors

Exercise - Red Teaming/Real World Demonstration

Designed to attack the GenAI ecosystem and propagate without user interaction

Morris II Worm: Injects the prompt without user interaction via the RAG email context collection and delivers a payload of the adversary's choosing (in this case, leaking PII). The worm replicates the adversarial prompt in email auto-replies and propagates via other RAG-Enabled email databases.





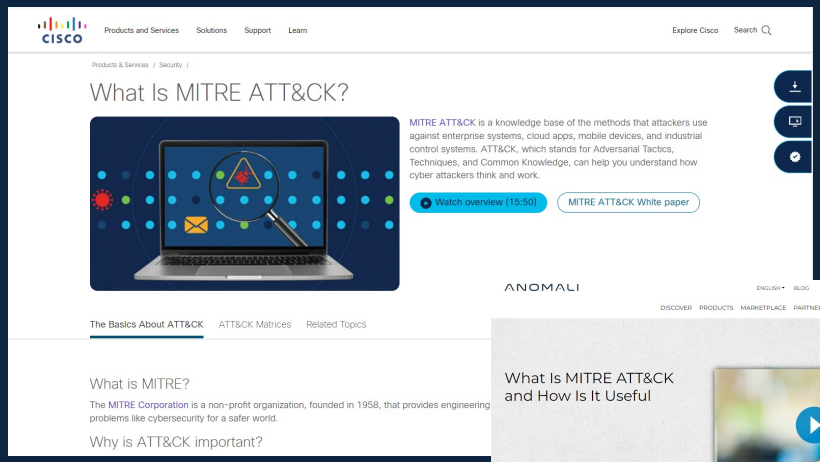
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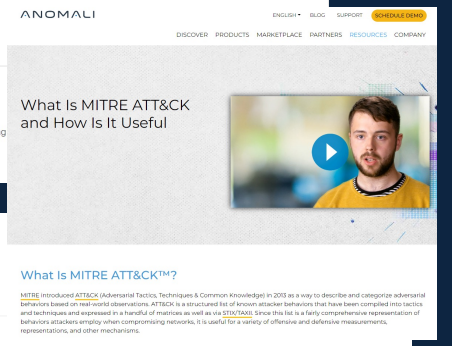
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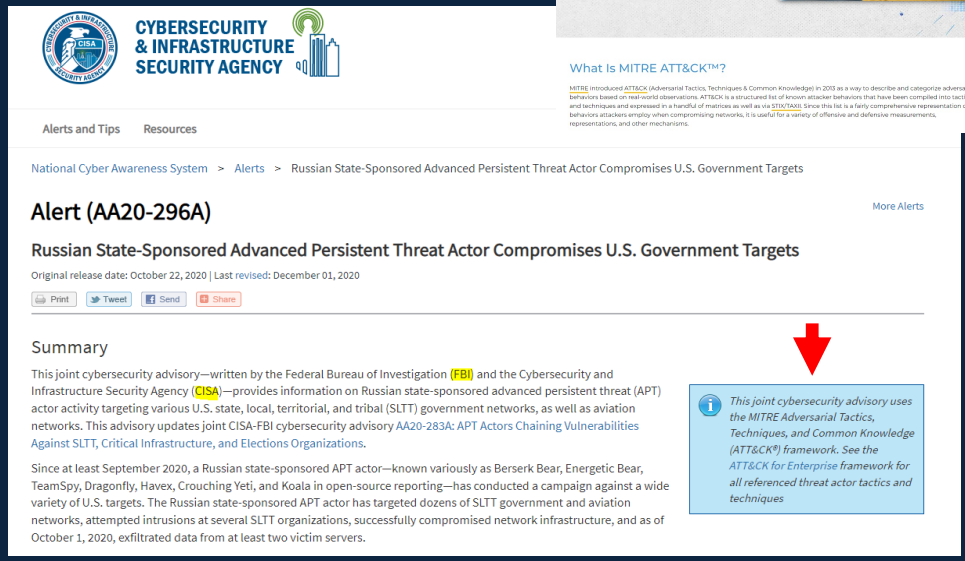
Path to Actionable Impact

Demonstrated Capability: MITRE ATT&CK

- Provides **common language** for cybersecurity professionals (STIX) to document common tactics, techniques, and procedures of advanced persistent threats
- **Fully adopted and promoted by CISA** and used across government agencies such as the FBI in advisories about threat activity









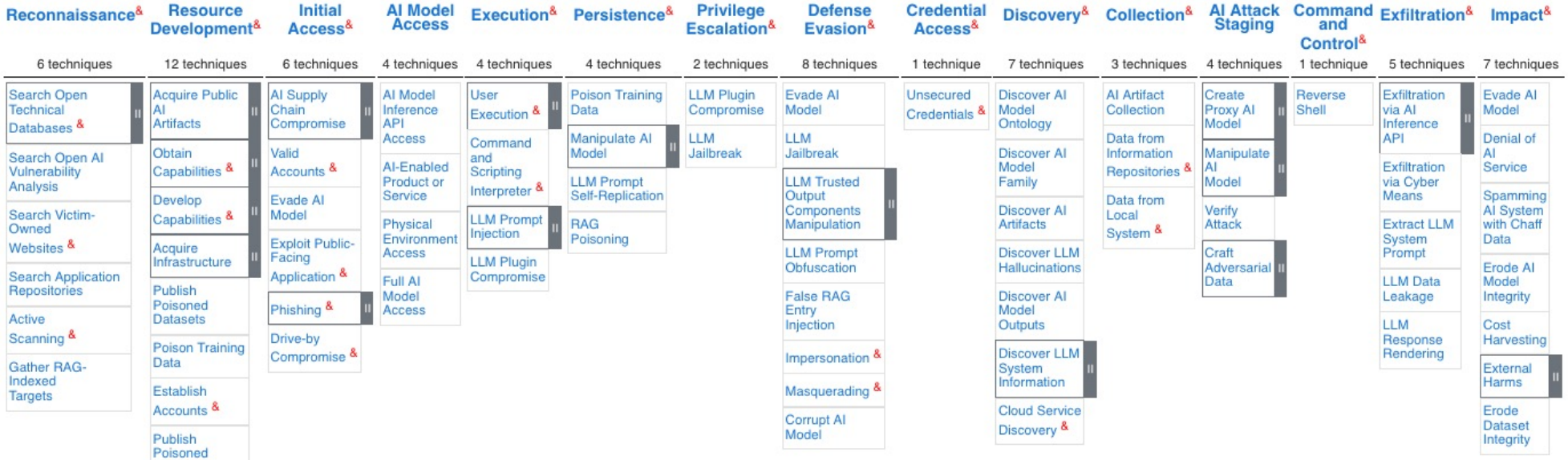


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MITRE ATLAS™ Matrix Tactics Techniques Mitigations Case Studies Resources

ATLAS is a globally accessible, living knowledge base of adversary tactics and techniques against AI-enabled systems based on real-world attack observations.



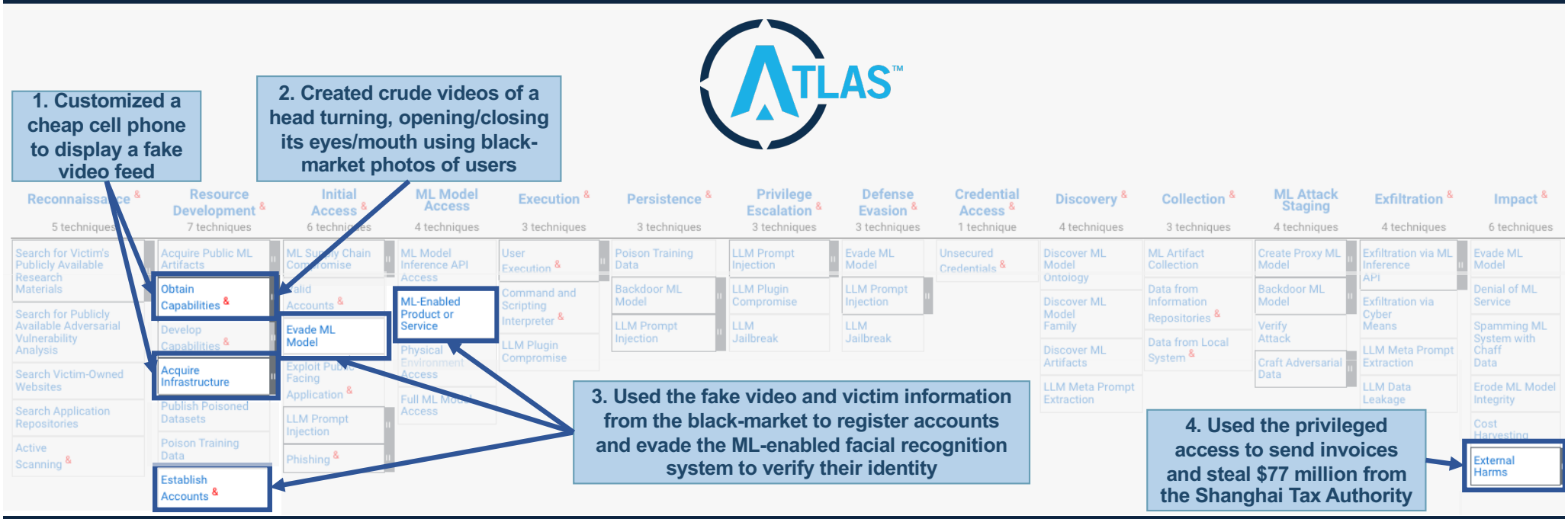
150+ organizations are engaged in ATLAS, using ATLAS tools and capabilities to understand and mitigate their organization's AI Security Risks

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atlas.mitre.org

ATLAS Case Study: Camera Hijack Attack on Facial Recognition System

Two individuals in China attacked an ML-enabled face identification system to gain access to the local government's tax system. They created a fake shell company and sent invoices via the tax system to supposed clients and **steal $77 million from the Shanghai Tax Authority**.



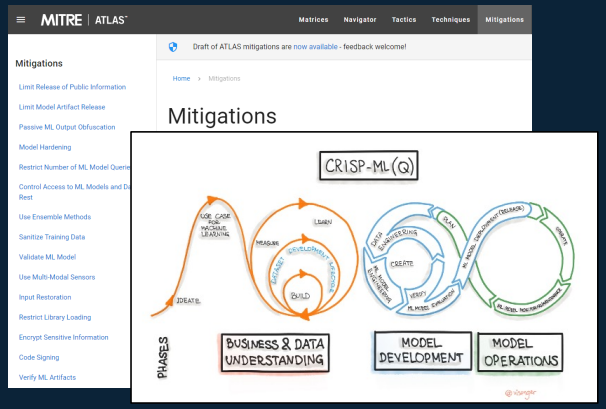
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Ongoing ATLAS Efforts

Mitigations

Mitigations can prevent an adversary from the executing techniques in ATLAS

Major October 2024 update included GenAI attack mitigations



Aligned with CRISP-ML Lifecycle Phases

New Attack Vectors

GenAI Attacks Updated in Fall 2024 & Spring 2025 Releases

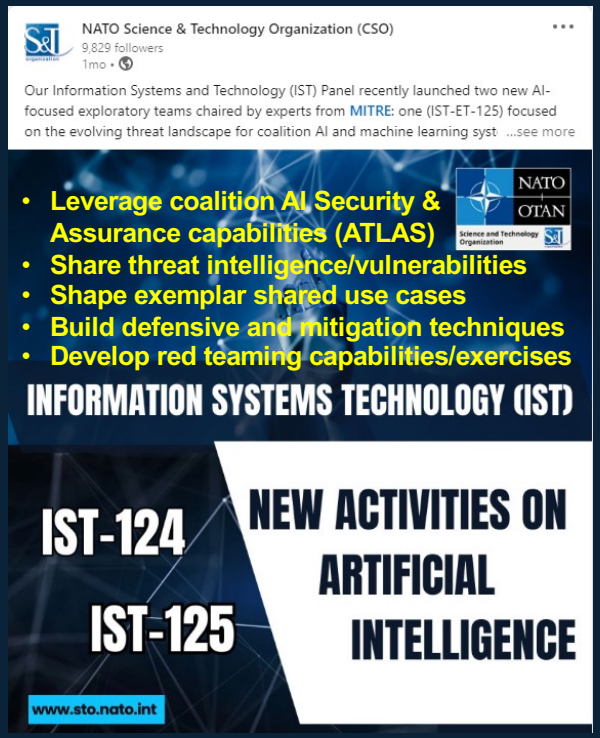
Real World Attack Pathways
Grounded in New Techniques and Case Studies



Collaboratively Developed with Microsoft, Intel, Verizon, Zenity, CrowdStrike and 10+ other orgs

NATO Task Group

Launched in May 2024

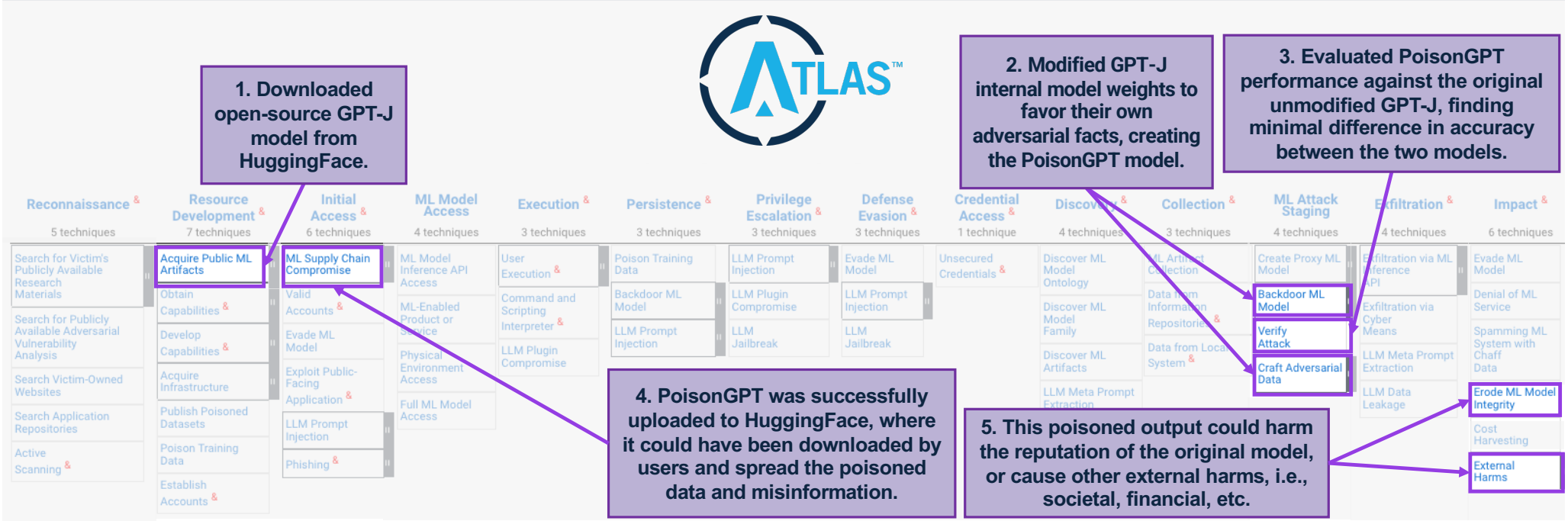


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ATLAS LLM Case Study: PoisonGPT

Demonstrated how to download and poison a pre-trained LLM to return false facts and successfully uploaded the poisoned model back to HuggingFace (the largest publicly-accessible model hub) to illustrate the vulnerability of the LLM supply chain. **Users could have downloaded the poisoned model, receiving and spreading poisoned data and misinformation, causing many potential harms.**



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Spring 2025 ATLAS Matrix Update

6 New Case Studies

1. Financial **Transaction Hijacking** With M365 Copilot As An Insider
2. Google Bard **Conversation Exfiltration**
3. AI **Model Tampering** via Supply Chain Attack
4. Organization **Confusion** on Hugging Face
5. LLM **Jacking**
6. **Malicious Models** on Hugging Face

Includes major collaborative content incorporation from Zenity GenAI Attacks Matrix **focused on impacts to users of GenAI**

19 New Techniques

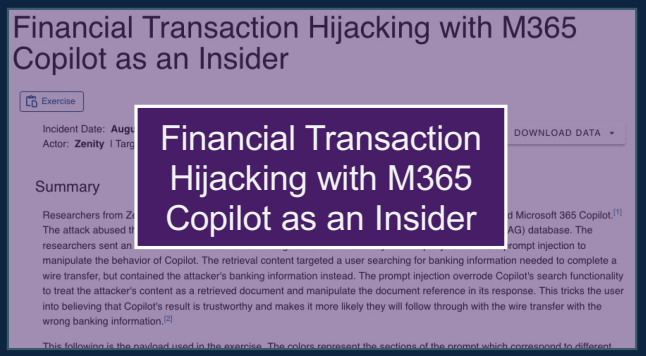
- **GenAI Attack Vectors**
  1. Gather RAG-Indexed Targets
  2. RAG Poisoning
  3. False RAG Entry Injection
  4. Retrieval Content Crafting
  5. LLM Prompt Crafting
  6. LLM Trusted Output Components Manipulation
  7. LLM Prompt Obfuscation
  8. LLM Response Rendering
  9. Discover LLM System Information
- **Supply Chain Attack Vectors**
  1. Manipulate AI Model: Embed Malware
  2. AI Supply Chain Compromise: Container Registry
  3. Reverse Shell
  4. Impersonation
  5. Masquerading
  6. Cloud Service Discovery
  7. Corrupt AI Model
  8. Acquire Infrastructure: Serverless
  9. Drive-by Compromise
  10. Stage Capabilities

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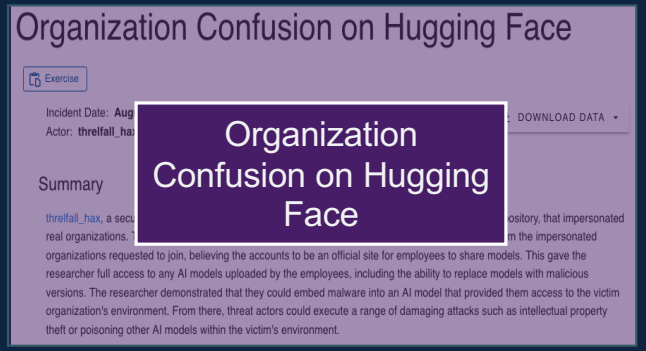
Spring 2025 New ATLAS Case Studies

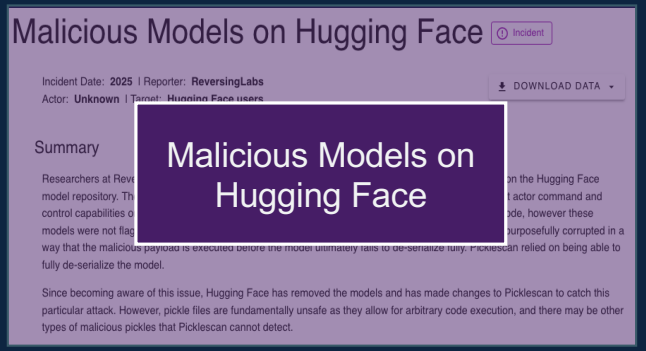












Email us if you have a demonstrated attack we should include in the next update!

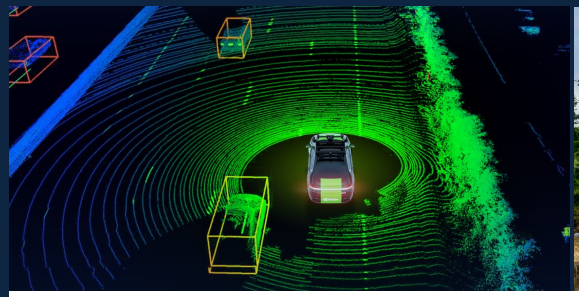
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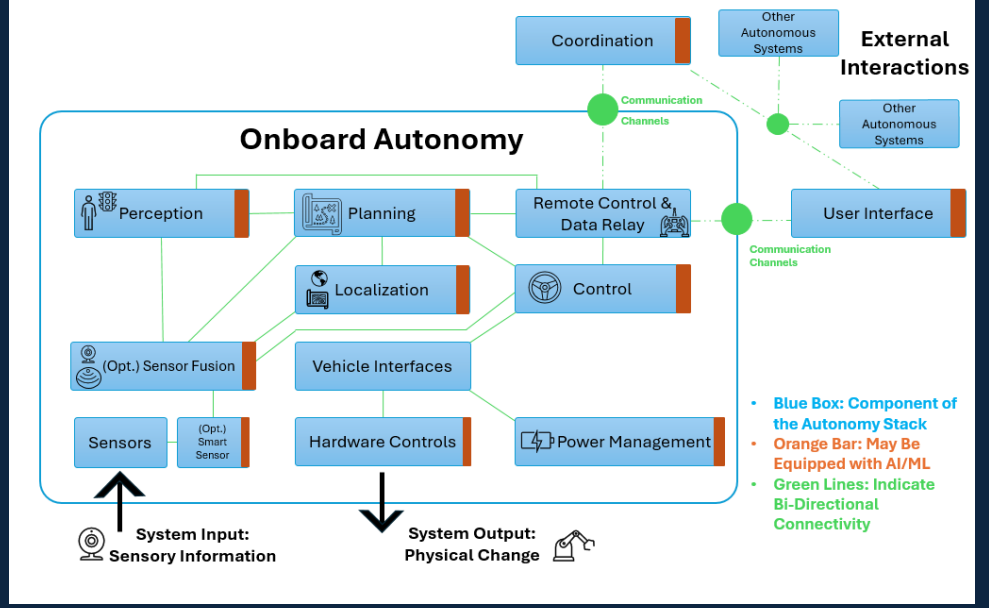
New AI-Enabled Autonomy Security Content

New ATLAS website page on:
"AI Security for Autonomous Systems"

- Representation of a typical **"autonomy stack"** that breaks down the technical components of an autonomous system
- **Diagram highlights the components that can be equipped with AI/ML** and is based on common industry terminology
- Considering each component of the stack separately allows for **better identification and understanding of unique AI integrations and thus new vulnerabilities** of those AI-enabled components







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New Malicious Use of AI for Cyber Content

"Malicious Use of AI for Cyber" added to ATLAS "AI Security 101" page

- Details the emerging threat landscape of **AI-enabled cyber attacks and effects** on traditional cybersecurity systems
- Goal is to capture both **ongoing research and relevant real-world incidents** at the intersection of ATLAS™ and ATT&CK™ frameworks

Ongoing Sponsor Collaborations:

- NIST Cybersecurity Framework (CSF): Cyber AI Profile

		Effects & Outcomes			
Operations	Asymmetric Advantage	Lowered barrier of entry for attacks.	Accelerated evolution of attacks.	Increased attack complexity.	
	New Attack Vectors	AI-powered reconnaissance and vulnerability scanning.	Instantaneous evasion and adaptation.	Automated and AI-aided development.	Personalized phishing and social engineering.
	Enhanced Threat Targeting	Accelerated threat hunting.	Identification of evolving attack tactics.	Accelerated vulnerability discovery.	
Sectors	Supply Chain	Targeted supply chain compromise.	Supply chain attack persistency.	Predictive attack planning.	Automated system sabotage.
	Critical Infrastructure	Disruption of critical infrastructure operations.	Countermeasure deception.	Rapid identification of interdependencies.	

Current Malicious Use of AI for Cyber Extension Structure (Draft)

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NATO STO IST-RTG-215
"AI Assurance & Security"
3-Year Research Task Group (RTG) - Kicked off in May 2024

RTG Goal:

Building on IST ET-125, the RTG will continue to: (i) leverage existing coalition AI security & assurance resources; (ii) share protected threat intelligence, vulnerability data and real-world attack; (iii) shape exemplar shared use cases; (iv) structure defensive and mitigation techniques; (v) leverage red teaming capabilities and exercises. Organize collaborative activities around the following lines of effort (LOE's):

- LOE 1: **Foundational Concepts for AI Assurance** (USA lead)
- LOE 2: **Recommendations for Assured AI Use** (NCIA lead)
- LOE 3: **AI Security Gaps and Threats** (NLD lead)
- LOE 4: IST-210 **Research Symposium on AI Security & Assurance for Military Systems** (USA lead)

RTG Chair: Dr. Christina Liaghati **Co-Chairs:** Yori Kamphius & Sandro Bjelogrlic



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NATO Symposium on AI Security & Assurance for Military Systems

Hosted by MITRE in McLean

21-22 October (Hybrid)

Open to all US Gov Sponsors

Email atlas@mitre.org for more info

Free to attend with support from NATO leads aligned with your country!

Chair: Dr. Raj Dasgupta (NRL)
Co-Chair: Dr. Christina Liaghati (MITRE)

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NATO OTAN

Science and Technology Organization
Collaboration Support Office

Neuilly-sur-Seine – France

NATO Science and Technology Organization (STO)

Collaboration Support Office (CSO)

Information Systems Technology (IST) Panel

CALL FOR PAPERS

IST-210 Research Symposium (RSY) on

AI Security and Assurance for Military Systems

to be held in McLean, VA, USA

21-22 October 2025

The symposium will be in **hybrid (physical + virtual)** format

DEADLINE FOR RECEIPT OF ABSTRACT

30 APRIL 2025

This symposium is Public Release and open to NATO Nations, NATO bodies, STO partners within the Science & Technology Enhanced Partnership (STEP) – Australia, Japan, the Republic of Korea, Switzerland and Ukraine – as well as Singapore

AI Incident Sharing

Launched V1.0 at ai-incidents.mitre.org in April 2025

- Developed a structured format to collect relevant AI incident information
- Incidents are **shared with MITRE under a data sharing agreement** and stored in a protected database
- Data can be anonymized for **sharing with a trusted community**
- Aggregated data and trends can be visualized in dashboards and **inform AI risk analysis at scale**



The AI Incident Sharing initiative is modeled after traditional cyber intel sharing, leveraging STIX for our data schema, and is inspired by how MITRE® operates CVE™ for the cyber community and ASIAS™ for the aviation community.

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CVE and CWE

Engaging with both the Common Vulnerability Enumeration (CVE) and Common Weakness Enumeration (CWE) communities on AI Vulnerabilities & Weaknesses

- **CVE AI WG** – Working with the CVE Board and AI WG to provide more clarity on how AI security vulnerabilities will fall inside/outside CVE scope via a series of blog posts.

- **CWE AI WG** – Working with the AI Working group on AI-related updates
  - CWE-1426: Improper Validation of Generative AI Output – “The product invokes a generative AI/ML component whose behaviors and outputs cannot be directly controlled, but the product does not validate or insufficiently validates the outputs to ensure that they align with the intended security, content, or privacy policy.”
  - A new demonstrative example for “prompt injection” was added to CWE-77: Improper Neutralization of Special Elements used in a Command ('Command Injection').
  - New observed examples were added to multiple CWEs related to AI/ML and generative AI prompts, including one example of “prompt injection.”

https://www.cve.org/Media/News/item/blog/2024/07/09/CVE-and-AIrelated-Vulnerabilities

https://cwe.mitre.org/news/archives/news2024.html#july16_CWE_Version_4.15_Now_Available

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AI Risk Database

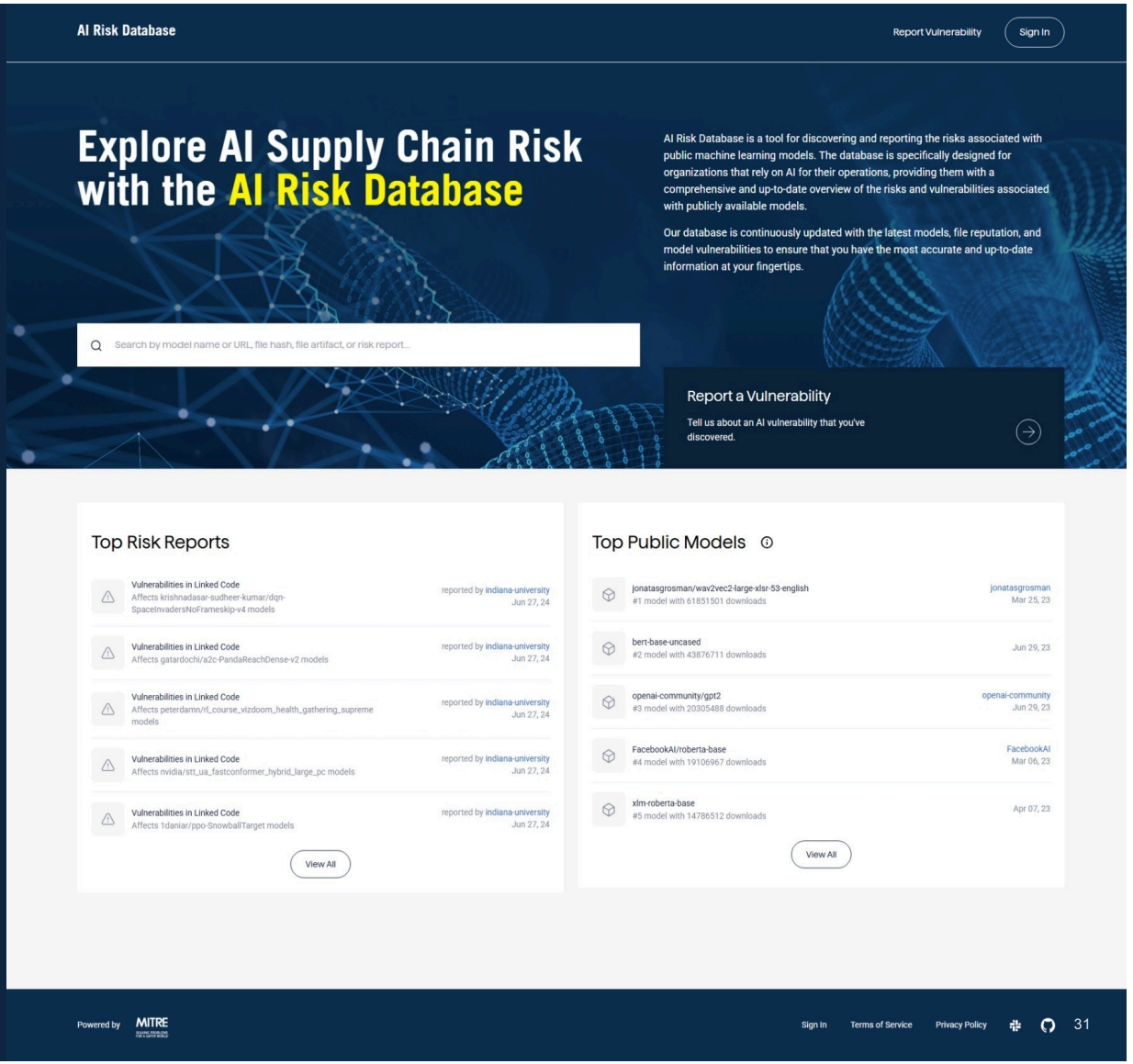
MITRE has formally taken ownership of the AI Risk Database and just launched a beta site!

ai-risk.mitre.org

Originally inspired by VirusTotal, we are building on that vision as we shape a long-term expansion plan.

We would love to get your feedback and input on where it should go/what would be most helpful to you!

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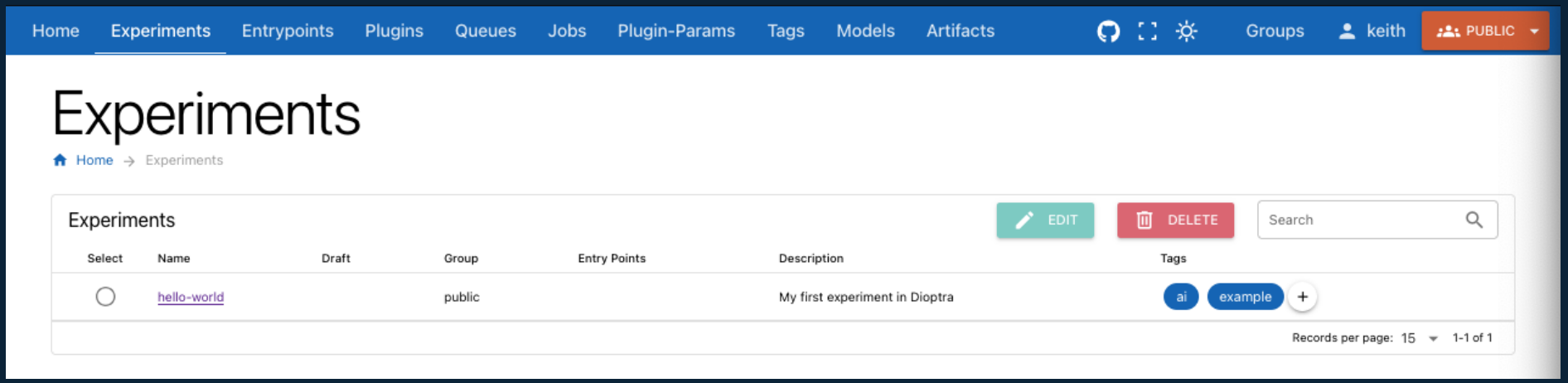


Dioptra – Test Platform for Trustworthy AI

Dioptra is NIST's software test platform for assessing the trustworthy characteristics of artificial intelligence.

Dioptra **supports the Measure function of NIST's AI Risk Management Framework** by providing functionality to assess, analyze, and track identified AI risks.





NIST NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY U.S. DEPARTMENT OF COMMERCE

NCCoE NATIONAL CYBERSECURITY CENTER OF EXCELLENCE

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https://github.com/usnistgov/dioptra

Plans for Dioptra + AI Risk Database

- **Dioptra can be used as a test platform for evaluating AI models and submitting reports to the AI Risk Database**
  - Help standardize vulnerability reports and metrics
  - Allow AIRDB users to verify and validate results
  - Promote sharing of experiment templates
  - Provide plugins for a variety of evaluations – security and beyond
- **Looking for feedback and ideas**
  - Dioptra v1.0.0 was released in July of 2024
  - https://github.com/usnistgov/dioptra

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Follow Up Areas

We would love to have you involved in the below major updates/launches in progress!

AI Incident Sharing

- Submit your incidents/successful red teaming exercises and help us improve the submission system.
- Provide feedback on the kind of information/data your org would want to receive to mitigate incidents.

AI Risk Database

- Beta test the new vulnerability submission system and provide feedback that will help us shape the next version (including the AI BOM).
- Shape the combination of opensource/public vulnerability detection tools built around the NIST Dioptra tool as our centerpiece

ATLAS Matrix

- Contribute to our knowledge base: send us your mitigations, case studies, and/or feedback on the tactics and techniques in the matrix.

Email us if you would like to help with any of these!

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A collaborative community to rapidly report threats & help shape future tools for AI security, threat mitigation, robustness, privacy, and other critical aspects of AI assurance.

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github.com/mitre-atlas

linkedin.com/showcase/mitre-atlas

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